PRAMOD HEGDE

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EXPERIENCE

Software Engineer Intern | Center of Competence - RVCE **Onsite | Oct - Dec 2024**
Personalized Medicine Recommendation System | Python, Flask, Scikit-learn, SVC, Pandas, NumPy, Pickle, Machine Learning, Feature Engineering, HTML, CSS, Bootstrap

- Developed a machine-learning-based medical recommendation web application using **Python, Flask, NumPy, Pandas, Scikit-learn,** and **HTML/Bootstrap**, enabling users to input multiple symptoms and receive a predicted disease.
- Integrated a trained **Support Vector Classifier** (SVC) using **Pickle**, converting user-provided symptoms into a binary feature vector across 130+ symptoms and mapping model predictions to corresponding disease classifications.
- Implemented a personalized recommendation pipeline that retrieves **disease-specific medications, precautions, dietary recommendations, workout suggestions, and medical descriptions** from structured **CSV datasets** using **Pandas**.
- Designed a responsive Bootstrap-based web interface with modal-based result sections for predicted disease, description, precautions, medications, diet, and workouts, integrating the ML prediction pipeline with the web application.

PROJECTS

LIBRACORE – Library Management System | Java, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, MySQL, Docker, Railway, Swagger/OpenAPI

- Developed a RESTful backend application using **Java 17, Spring Boot, layered architecture, REST APIs, Controllers, Services Repositories, DTOs,** and **Entity mapping** for library management.
- Implemented **JWT-based stateless authentication** and **role-based access control (RBAC)** using **Spring Security**, supporting **ADMIN** and **MEMBER** roles with protected endpoints.
- Built **CRUD REST APIs**, pagination using Spring Data, and book inventory management with **JPA/Hibernate**, including automatic Available Copies updates during book issue and return transactions.
- Integrated **MySQL** using **Spring Data JPA/Hibernate,** implemented custom exception handling and validation**,** and developed integration and security tests for **authentication, authorization,** and core business workflows.
- Containerized the application using **Docker** and **Docker Compose**, configured environment variables for production, documented APIs using **Swagger/ OpenAPI**, and deployed the production backend on **Railway** with **HTTPS**.

PE & URL Malware Detection |Python, Streamlit, Scikit-learn, TensorFlow/Keras, Pandas, NumPy, PEfile, Joblib, VirusTotal API, REST API, Docker, Machine Learning, Feature Engineering

- Developed an **end-to-end malware detection application** using **Python, Streamlit, Scikit-learn, TensorFlow, and Pandas**, supporting both **Windows PE file analysis and malicious URL detection** through machine-learning models.
- Implemented **PE malware detection** by parsing EXE/DLL/SYS files with **PEfile**, extracting **PE header, section, import/export, entropy, and resource features**, and enforcing consistent feature ordering before model inference.
- Integrated a trained **Random Forest classifier and feature scaler** using **Joblib and Scikit-learn** to classify PE files as **malicious or legitimate**, with prediction probabilities, confidence scores, and extracted technical characteristics displayed through an interactive **Streamlit** interface.
- Built a **malicious URL analysis pipeline** using **TensorFlow/Keras**, URL feature extraction, input validation, and **VirusTotal API integration** to submit URLs for scanning and retrieve security-analysis results from multiple detection engines.
- **Containerized and deployed** the application using **Docker**, configured the **Streamlit** service for production execution, managed Python dependencies and ML model artifacts, and exposed an interactive security-analysis dashboard for real-time malware detection.

TECHNICAL SKILLS

Programming Languages: Python, Java

Backend & Frameworks: Spring Boot, Spring Security, Django, REST APIs, JWT, JPA, Hibernate

Databases: SQL, MySQL, MongoDB

DevOps & Deployment: Docker, Docker Compose, Railway, Render

API & Documentation: Swagger, OpenAPI, API Integration

Frontend: HTML, CSS, Bootstrap

Tools & Version Control: Git, GitHub, VS Code, PyCharm, Intellij

EDUCATION

Master of Computer Applications (MCA), R V College of Engineering, Bangalore, CGPA: 7.72, 2023–2025.

Bachelor of Computer Applications (BCA), JSS SMI UG C PG Studies, Dharwad, CGPA: 7.27, 2020–2023.